Military Specification

Microphone, Carbon, Oxygen Mask, ANB-M-C1

This Specification has been approved by the
Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 This specification covers Microphone, Carbon, Oxygen Mask, ANB-M-C1 for use in military equipment.

1.2 Classification - The microphone shall be of one type and shall consist of the following.

Unit	Type Designation	Max. Allowable Weight Ounces	Appl. Para.
Microphone, Carbon, Oxygen Mask	ANB-M-C1	1.5	3.4.1.2

1.3 Microphone, Carbon, Oxygen Mask, ANB-M-C1 - The microphone shall consist of the following items:

(1) Transmitter MC-254-A; see 3.4.1.2(1)
(2) Cord assembly; see 3.4 1.2(2)
(3) Connector, type PJ-292; see 3.4.1.2(3)

When a complete microphone is not required, any one item or combination of items may be procured as specified in the procurement document (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein

SPECIFICATION

Military

MIL-P-642	Plugs, Telephone
MIL-C-5186	Cordage CO-219
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-P-17555	Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts

FSC-5965

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2.2 Availability of Documents -

(1) When requesting applicable documents give both the title and number. All requests shall be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (CDS), Philadelphia 11, Pennsylvania.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Deviations from this specification, or from subsidiary documents where applicable, may be made only upon written approval.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Qualification - The microphones furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern, except as otherwise specified herein.

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specification.

(3) When a non-standard part is used, the meeting of general material requirements such as non-flammable, fungus resistant, non-toxic, etc., as required by Specification MIL-E-5400 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

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3.3 Design and Construction -

3.3.1 General - The microphone ANB-M-C1, when installed in a suitable mask and connected with suitable intercommunicating or communications equipment, shall convert acoustic energy into electrical energy and thereby affect the transmission of clear speech.

3.3.2 Design - The detailed mechanical and electrical design of this microphone unit shall be accomplished by the contractor subject to the requirements of this specification, these requirements being detailed herein only to the extent considered necessary to obtain the desired mechanical and electrical characteristics and performance, and permanence of the same. The microphone unit shall provide the specified performance when tested in the manner detailed along with the respective performance requirements and shall provide usable performance when used in any position (horizontal, vertical, etc).

3.3.3 Service Conditions - The microphone shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment with the following exceptions.

3.3.3.1 Altitude - The microphone shall be designed to reduce the effects of changes in altitude. At 40,000 feet, the frequency response shall be substantially unchanged from the response at ground level.

3.3.4 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the microphone.

Temperature	Room ambient 25°C (+15°C or -5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity

3.3.5 Weight - The weight of the microphone ANB-M-C1 shall not be greater than 1.5 ounces.

3.3.6 Marking - Each microphone ANB-M-C1 and each transmitter assembly MC-254-A, P/O ANB-M-C1, shall be permanently and legibly marked as shown in Figures 1 and 2. The manufacturer's trade mark may be added on the transmitter unit, if desired

3.3.7 Pressure Equalization - Suitable provision shall be made in the design to equalize the pressures in and on all sections of the microphone unit, when it is subjected to a change in altitude.

3.3.8 Carbon Granules - Carbon granules shall be retained in the microphone unit in such a manner that it is impossible for them to leak from the unit under conditions of normal service use.

3.3.9 Interchangeability - The interchangeability requirements as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units with the latest Model of any previously produced units. Such interchangeability shall be measured against a model, manufacturing drawings or other technical information as provided by the Bureau of Aeronautics. The lack of such material shall not, in any way, constitute a waiver of this requirement In the event that such material is not provided as government furnished under any given procurement, the contractor is required to submit a formal request for guidance to the Bureau of Aeronautics.

3.4 Detail Requirements -

3.4.1 Microphone, Carbon, Oxygen Mask, ANB-M-C1 - The microphone assembly shall consist of the following:

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3.4.1.1 Form Factor - The microphone assembly shall not weigh in excess of 1.5 ounces and shall have dimensions shown in Figure 1 ,

3.4.1.2 Contents of the Unit - The microphone shall be composed of the following:

(1) Transmitter Assembly, MC-254-A, P/O ANB-M-C1, as shown in Figure 2, shall be individually interchangeable with components previously manufactured and shall not weigh in excess of 1.0 ounce.

(2) Cord Assembly - The cord shall be non-retractile, shall have two conductors and shall be type CO-219 in accordance with Specification MIL-C-5186 or equivalent. Dimensions and marking shall be as shown in Figure 3.

(3) Connector - A type PJ-292 plug in accordance with Specification MIL-P-642 shall be provided for the cord assembly.

3.4.1.3 Anchor - The cord shall be anchored to withstand a tension of 20 pounds at PJ-292 Plug with cord connections not completed.

3.5 Performance -

3.5.1 Response Characteristics - Shall be measured over the audio frequency range from 300 to 4000 c.p.s. A sound pressure of 115 db. above a reference level of 0.00020 dyne per sq. cm shall be applied at the grid of the microphone at the test frequencies indicated in the following Tables I, II, and III to determine adequately the microphone response. The response curve shall lie within the limits of the two tolerance lines shown on Figures 4 or 5 or 6, which are plotted in accordance with the frequencies indicated on Tables I, II and III respectively.

TABLE I

Frequency	Db relative to 1 Millivolt for Constant Sound Pressure of 115 db. Ref. 000200 Dynes/Sq Cm.		
	Minimum	Spread	Maximum
300			27
400			33
500	26	11	37
600	32	10	42
700*	36	9	45
800	39	9	48
900	42	9	51
1000	46	9	55
1200*	51	9	60
1400	54	7	61
1600	55	7	62
1800*	56	7	63
2000	57	7	64
2200	58	6	64
2400*	59	6	65
2600	59	7	66
2800*	57	7	64
3000	55	8	63
3200	53	8	61
3400*	51	9	60
3600	50	9	59
3800			58
4000			57

Frequencies marked '*' to be used in production tests on product.

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TABLE II

Frequency	Db relative to 1 Millivolt for Constant Sound Pressure of 115 db. Ref. .000200 Dynes/SQ. Cm.	
	Minimum	Maximum
300	-	+23
500	+22	+30
1000	+42	+52
1200	+46	+55
1500	+51	+59
2000	+55	+63
2500	+58	+66
3000	+56	+64
3500	+51	+62
4000	-	+60

TABLE III

Frequency	Db relative to 1 Millivolt for Constant Sound Pressure of 115 db. Ref. .000200 Dynes/Sq. Cm.	
	Minimum	Maximum
300	-	+24
500	+25	+36
1000	+44	+53
1200	+47	+55
1500	+50	+58
2000	+52	+60
2500	+56	+64
3000	+56	+64
3500	+52	+62
4000	-	+60

The response curve of the microphone unit shall be measured in the test apparatus and circuit shown on Figures 7 and 8. The response of the microphone shall fall off as rapidly as practicable below 300 c p. s. and above approximately 4000 c p. s. In any production testing of both individual and representative samples, a tolerance of one Db. shall be permitted beyond the limits specified by Tables I, II and III to allow for the recognized daily variation in the test equipments.

3. 5. 2 Response Curves - Shall be plotted on twenty division per inch by two 4-1/2 inch cycles semi-log graph paper. Graphs should be plotted as voltage developed across the 100 ohm load resistor (in decibels with respect to one millivolt) versus frequency for the sound level of 115 db. with respect to 0. 00020 dynes per sq. cm. The ordinates of the paper should read from +10 to +70 db. with respect to 1 millivolt.

3 5. 3 Altitude Operation - The microphone shall be designed to reduce the effects of changes in altitudes. At 40,000 feet, the frequency response shall be substantially unchanged

3 5. 4 Anti-Packing Characteristics - Microphones shall be designed to transmit intelligible speech over continuous periods of six hours under overload conditions of 60 ma. button current without adverse effect upon microphone operation. In addition, the anti-packing characteristics shall not be seriously affected when operating the microphone in positions other than the vertical position.

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4.3.2.1 Sampling Instructions - One microphone from the first 10 production microphones, and thereafter two from each lot of 100, or fraction thereof, as selected by the cognizant Government Inspector shall be given the specified sampling tests.

4.4 Tests Required - The microphone sample shall be subjected to the tests listed in Table IV and any other tests deemed necessary by the Bureau of Aeronautics or the cognizant Government Inspector to determine compliance of the microphones with the requirements of this specification.

TABLE IV

Tests	Test Procedure	Qualification Tests	Individual Tests	Sampling Tests
Examination of Product	4.5.2	X	X	X
Response Measurement	4.5.3	X	X	X
Cord Test	4.5.4	X		
Humidity	4.5.7	X		X
Temperature Shock	4.5.8	X		X
High Temperature	4.5.9	X		*
Low Temperature	4.5.10	X		*
Pressure Cycling	4.5.5	X		X
Pressure Equalization	4.5.6	X		X
Vibration	4.5.11	X		*
Shock	4.5.12	X		*

* These tests shall be performed on the sample microphone from the first 10 production units and on the sample units for every fifth lot

4.5 Test Methods -

4.5.1 Test Circuit Apparatus - The test circuit apparatus shall be assembled and connected in a manner similar to that shown in Figures 7 and 8 with provision for the entire stand to be rotated through 360 degrees in a vertical plane which is perpendicular to the face of the microphone.

4.5.1.1 Audio Oscillator - The Audio Oscillator shall be of good design, having a wave form distortion of 2 per cent or less, and with an associated amplifier of similar quality having an output adequate to operate a W. E. 555, or equivalent receiver unit up to a level of 115 db, above .00020 dynes per sq. cm. (A General Radio Type 713B Beat Frequency Oscillator, or its equivalent is recommended.)

4.5.1.2 Vacuum Tube Voltmeter - The two vacuum tube voltmeters used in the equipment must have a flat frequency response from at least 100 cycles to 10,000 cycles, and must be capable of measuring voltages from .001 volts r.m.s. to 10 volts, or more (Ballantine Model 300 vacuum tube voltmeters or equivalent are suggested for this application).

4.5.1.3 Sound Source - The sound source should be a W. E. 555 receiver unit or equivalent It is generally intended that the distortion in the output of the sound source shall be such that the second harmonic is at least 35 db, the third at least 40 db, and the fourth at least 45 db, below the fundamental. The W. E 555 receiver unit has a voice coil impedance of about 15 ohms. Suitable field coil excitation (1.5 amperes d.c. ±10%) should be provided.

4.5.1.4 Baffle - The baffle for W. E 555 Driver Unit - a baffle (wooden ring of Figure 7 shall be mounted on the front of the W E. 555 unit. The front face of the baffle shall be in a plane parallel to the plane of the face of the microphone baffle

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4. 5. 1. 5 Coupler - The coupler shown on Figure 10shall be mounted so that 1/4 inch air space exists between the extreme end of the coupler and the front plane of the microphone baffle (Figure 9). The test circuit shown in Figure 8 shall be used. When the switch S. C. is closed, the value of R_1 is set so that a current of 60 milliamperes will flow through the meter. The AC voltage developed across the resistor R_2 should be measured by a Ballantine vacuum tube voltmeter, or equivalent These voltages expressed in Db relative to 1 millivolt shall be recorded and plotted as a function of frequency.

4. 5. 1. 6 Base - The test equipment shall be mounted on a rigid test base of suitable design, with provision for rocking the carbon microphone units mounting baffle. A suggested model of this equipment is shown on Figure 7.

4. 5. 2 Examination of Product - Each microphone shall be examined carefully to determine conformance with this specification with respect to material, weight, dimensions, construction and workmanship.

4. 5. 3 Response Measurement Test - The response measurements and the conditioning of the microphone unit prior to test shall be as follows: A standard 640-A or equivalent condenser microphone and its associated pre-amplifier, shown on Figure 11 or equivalent shall be mounted in the baffle by means of the simulated microphone case shown in Figure 12. The electrical input to the artificial voice shall be so determined that the sound level produced at the grid of the microphone is 115 Db. with respect to a reference level of 0 00020 dyne per sq cm as measured by the W. E. 640-A condenser microphone at frequencies of 100 to 5000 c. p. s. This determination shall be made for each position in which the microphone is to be tested and need be made only as often as necessary to prove that no changes have taken place in the output of the W. E. 555 unit. Calibration shall be checked at all positions to be used in testing. Having determined the electrical input to yield a constant sound pressure level at the grid of the microphone, remove the standard microphone and the simulated case and mount the microphone to be tested in its baffle as shown in Figure 7. It is desirable to shake the microphone vigorously in all directions before mounting it in the test fixture. With the face of the microphone in the vertical plane, rotate the microphone unit and the fixture about the axis of the microphone unit, two times, through an arc of approximately one complete cycle of rotation (270 degrees and back) in two seconds. Then rotate the whole stand to the position in which the microphone is to be tested. Open switch S C. and wait approximately three seconds for current stabilization. Apply the agitating tone by sweeping the oscillator through the 1000-3000-1000 c. p. s. audio frequency range, three times, at a rate of approximately one complete sweep (1000-3000-1000) per two seconds. The driver input shall be controlled in such a manner that the maximum sound pressure applied to the microphone unit during the application of the sweep band shall be 127 db. At the conclusion of the third sweep cycle the audio oscillator shall be shut off in such a manner as not to destroy the previous conditioning. (There should be no loud clicks.) Turn on the test tone after one or two seconds delay and take the response curve point by point in accordance with paragraph 3. 5. 1 headed "Response Characteristics" starting at 300 cycles and extending to 4000 cycles, in sufficient detail to definitely establish the shape of the curve. It is necessary that the total time for taking the response curve does not exceed three minutes, in order that the carbon button does not completely lose its conditioning. The 640-A condenser microphone shall have been calibrated by the reciprocity method by one of the following laboratories:

National Bureau of Standards
Cruft Laboratory, Harvard University
Bell Telephone Laboratory

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4.5.4 Cord Test - The Cord shall be tested in accordance with Specification MIL-C-5186 to determine conformance with 3.4.1.2(2).

4.5.5 Pressure Cycling - The microphone unit shall be subjected to not less than 5 varying pressure cycles. Each pressure cycle shall consist of 30 minutes at 5 inches of mercury and 30 minutes at 30 inches of mercury. The pressure transition shall be approximately 5000 feet per minute. Upon completion of this test the response characteristics shall meet the requirements of this specification (3.5.1).

4.5.6 Pressure Equilization - The microphone shall be placed in an altitude chamber and the pressure inside the chamber reduced to that corresponding to an altitude of 15,000 feet (approximately 16.7 inches of mercury) The response of the microphone shall be measured while the pressure is varied at a rate of 5000 feet per minute (up or down) and shall remain within the limits specified in 3.5.1.

4.5.7 Humidity - The microphone shall be subjected to the humidity test procedure I of Specification MIL-E-5272. Upon completion of this test the response characteristics shall meet requirements of this specification, (3.5.1).

4.5.8 Temperature Shock - The microphone unit shall successfully meet the response characteristics after being subjected to the Temperature Shock Test Procedure II of MIL-E-5272.

4.5.9 High Temperature Test - The microphone shall be tested in accordance with Procedure I of MIL-E-5272.

4.5.10 Low Temperature - The microphone shall be tested in accordance with Procedure I of MIL-E-5272.

4.5.11 Vibration - The microphone shall operate satisfactorily after being subjected to the vibration test Procedure I of MIL-E-5272.

4 5.12 Shock - The microphone shall operate satisfactorily after being subjected to the Shock Test Procedure I of MIL-E-5272.

4.6 Pre-Submission Testing - No item, part or complete microphone unit shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest -

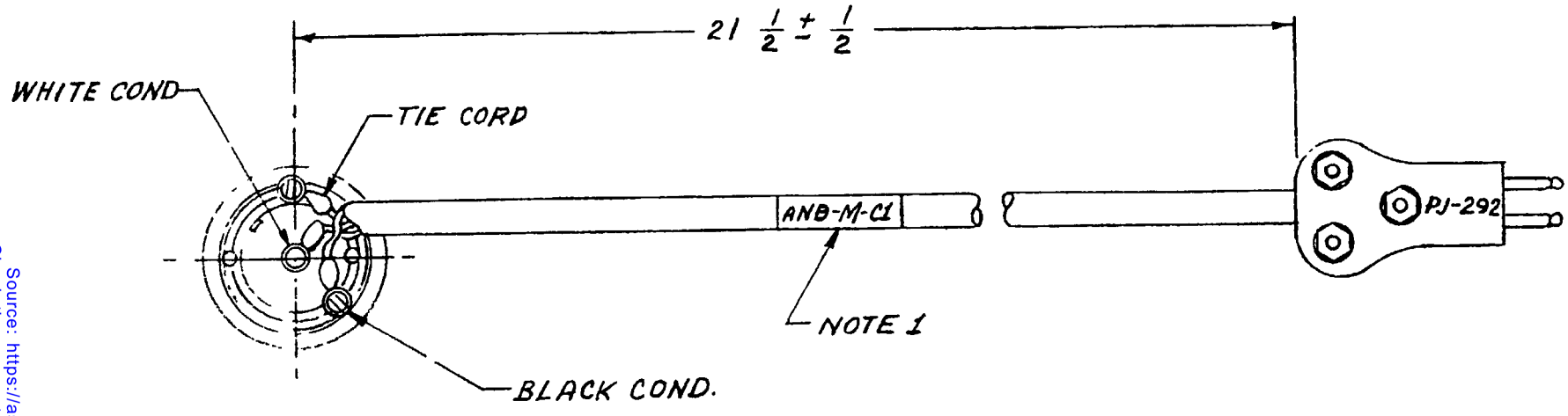
4.7 1 Individual Tests - Each microphone which fails to meet all requirements set forth in 4.3 1 shall require immediate rejection and may be reworked as necessary to meet all the requirements of this specification.

4 7.2 Sampling Tests - If any microphone unit selected fails to successfully pass all sampling tests, it shall cause rejection of the lot represented until all units have been reworked as necessary to correct defects Sampling of reworked microphone units for retest shall be made by the cognizant Government Inspector as necessary to insure that all defects have been corrected.

4 7 3 Reworked Units - The reworked microphone units which meet all requirements during retests may be accepted for delivery but complete details concerning the cause of failure and steps taken to insure correction of such causes in future production shall be forwarded by the cognizant Government Inspector to the Bureau of Aeronautics.

4.7.3.1 Failures - Should samples selected from any lot of reworked microphone units fail to meet all requirements during retest, the lots of microphones represented shall not be further reworked and submitted for further test without specific approval in writing from the Bureau of Aeronautics.

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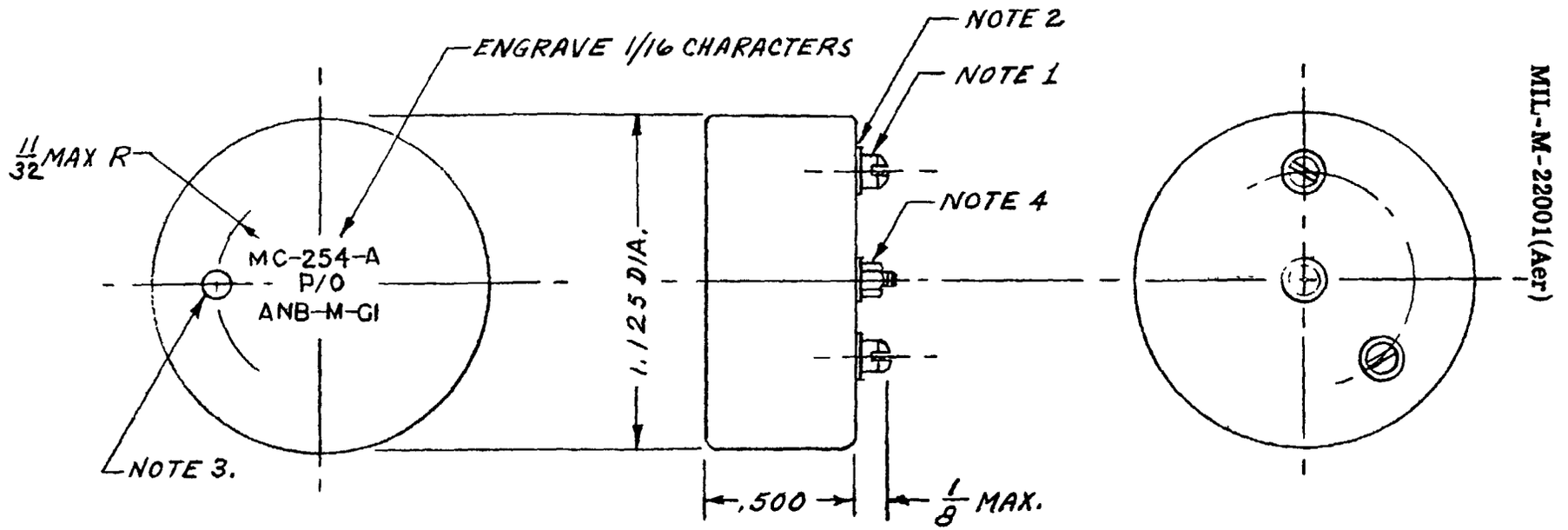


NOTES:

1. MARKER SHALL BE PERMANENTLY AND LEGIBLY MARKED ANB-M-C1 AND MAY INCLUDE THE MFG.RS PREFIX LETTERS.
2. TOLERANCES: FRACTIONS ± 1/64; DECIMALS ± .005 UNLESS OTHERWISE INDICATED.
3. DIMENSIONS ARE IN INCHES.

FIGURE 1. ANB-M-C1, ASSEMBLY

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NOTES: 1. SCREW, FILISTER HEAD, BRASS, ZINC PLATE, #2-56 X 3/32 LONG.

2. LOCK WASHER #1202, STEEL, ZINC PLATE AS MADE BY SHAKEPROOF INC. CHICAGO, ILL. OR EQUAL. 3 REQUIRED.

3. SIZE, NUMBER AND LOCATION OF HOLES IN CAP TO BE DETERMINED BY MANUFACTURER SUBJECT TO PERFORMANCE REQUIREMENTS AND LIMITING DIMENSIONS.

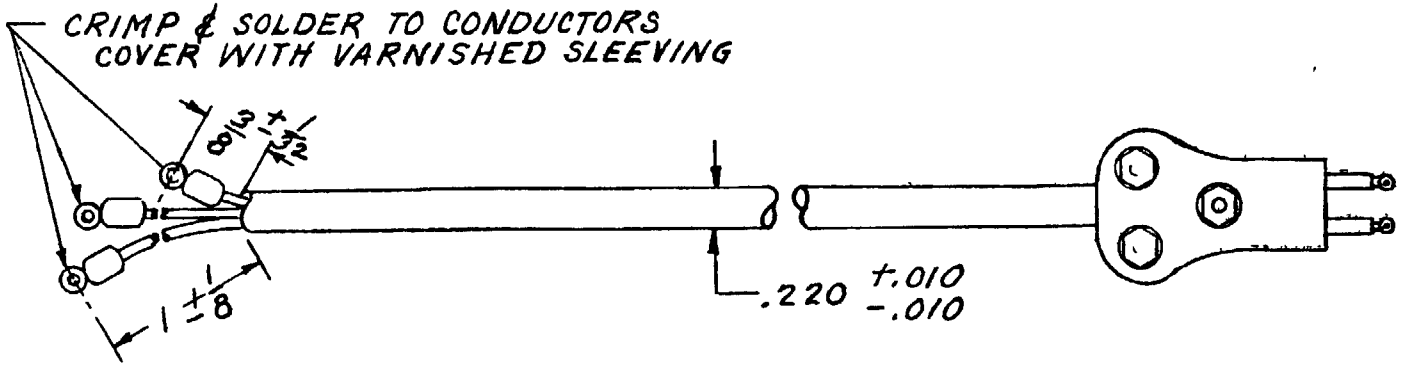
4. AT THE OPTION OF THE MANUFACTURER, CENTRAL TERMINAL MAY BE FASTENED BY NUT, HEX. HEAD, #2-56.

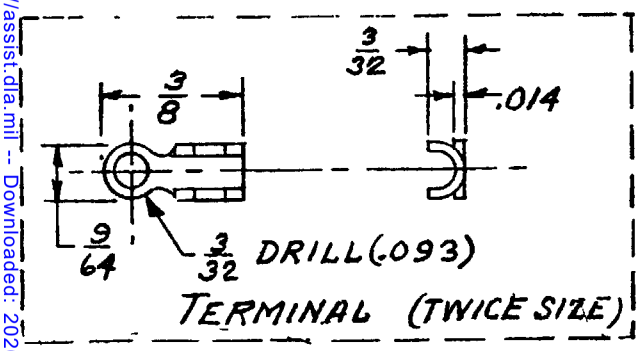
5. TOLERANCES: FRACTIONS ± 1/64; DECIMALS ± .005 UNLESS OTHERWISE INDICATED.

6. ALL DIMENSIONS ARE IN INCHES.

FIGURE 2. P/O ANB-M-C1 MICROPHONE ASSEMBLY

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NOTES:

1. TWO TERMINALS TM-89 AND ONE TERMINAL W.E. #120 AS SUPPLIED TO USAF OR ASO STOCK No. R17T4392-10 TERMINAL MAY BE USED IN LIEU OF TERMINALS SHOWN.
2. TOLERANCES: FRACTIONS ± 1/64 ; DECIMALS ± .005 UNLESS OTHERWISE INDICATED.
3. ALL DIMENSION ARE IN INCHES.

FIGURE 3. P/O ANB-M-C1, SUB-ASSEMBLY.

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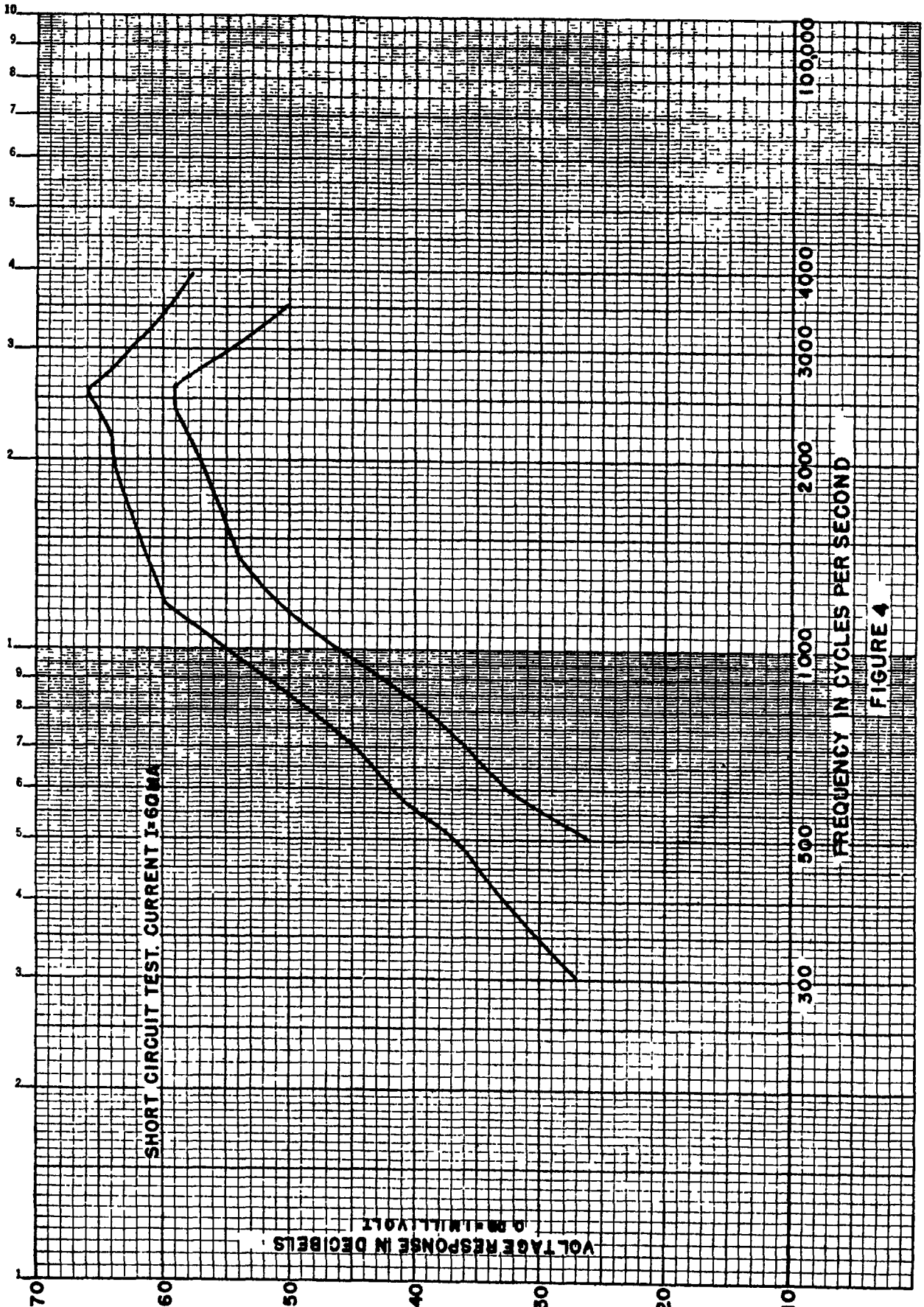
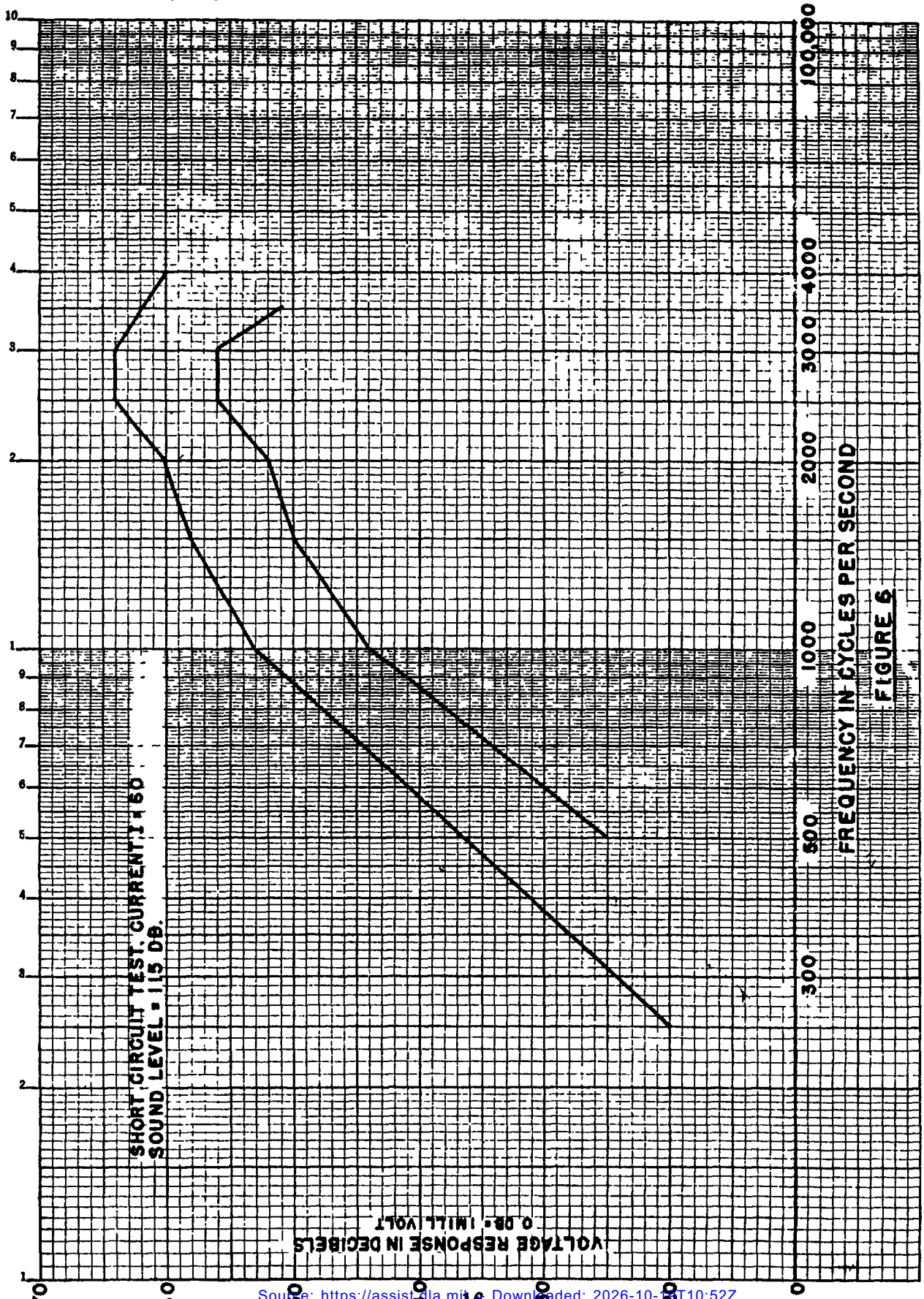
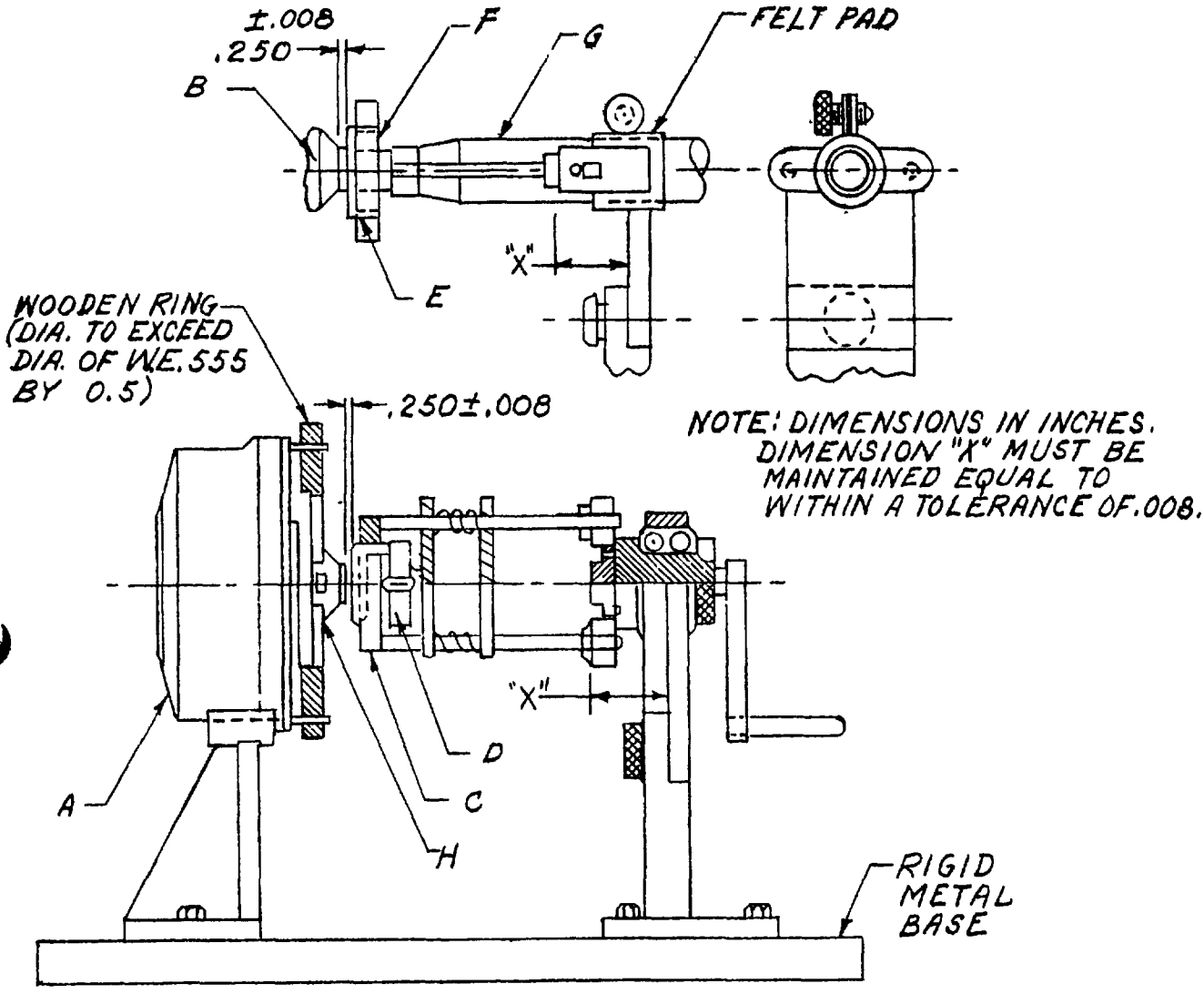


FIGURE 4

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A. W.E. 555 RECEIVER
B. MICROPHONE CALIBRATING COUPLER
C. MICROPHONE BAFFLE
D. MICROPHONE
E. MICROPHONE CALIBRATING BAFFLE
F. W.E. 640-A CONDENSER MICROPHONE WITH GRID.
G. PRE-AMPLIFIER
H. DRIVER UNIT COUPLER

FIGURE 7.

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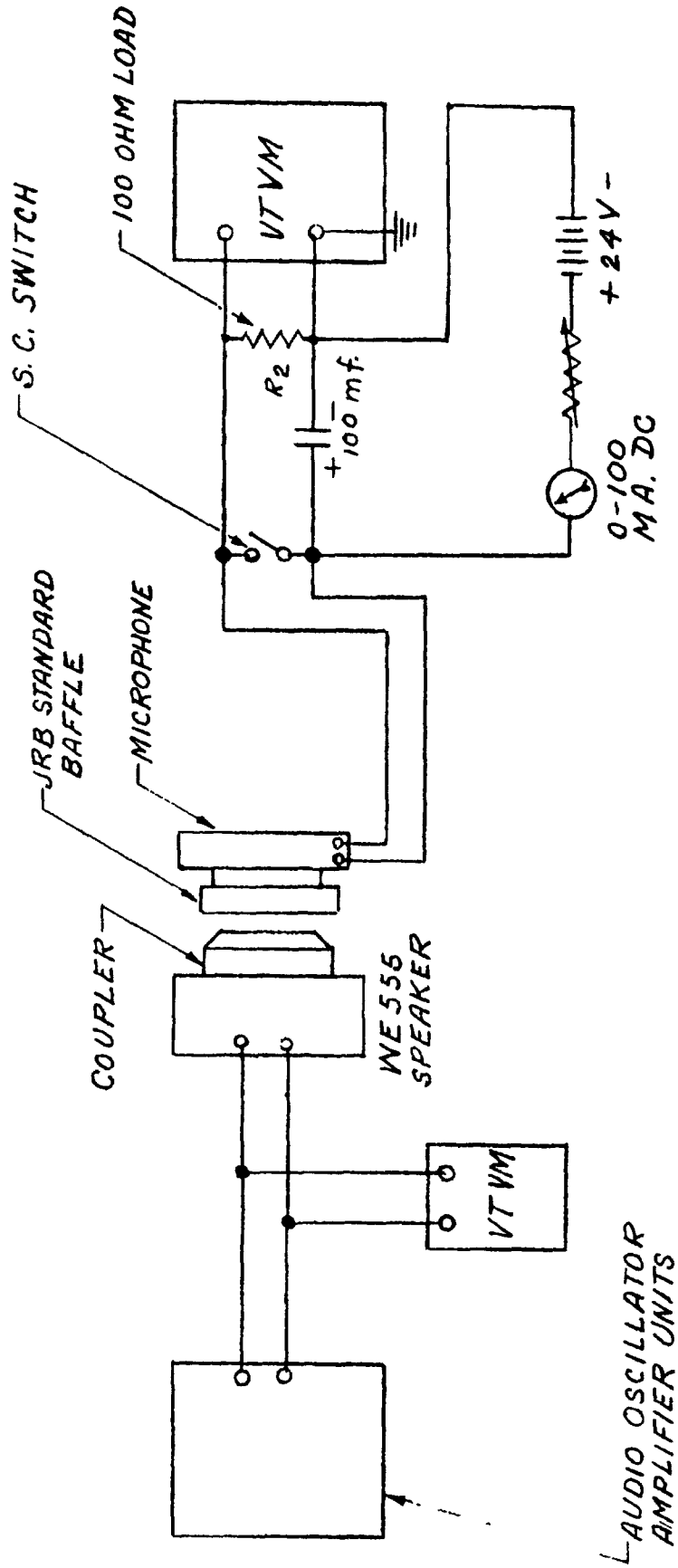
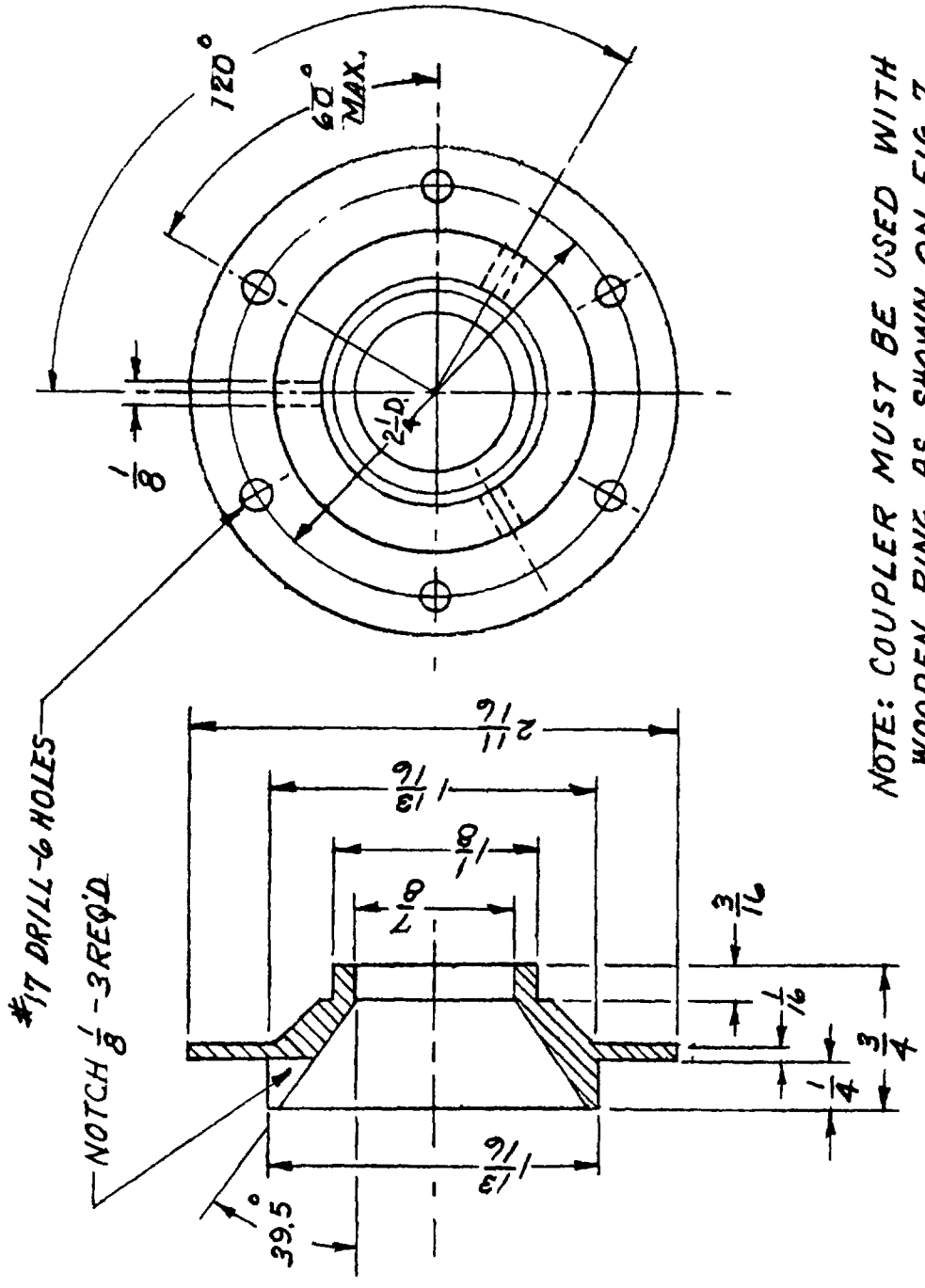


FIGURE 8. TEST CIRCUIT

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NOTE: COUPLER MUST BE USED WITH WOODEN RING AS SHOWN ON FIG. 7.
DIMENSIONS ARE IN INCHES.

FIGURE 10. DRIVER UNIT COUPLER

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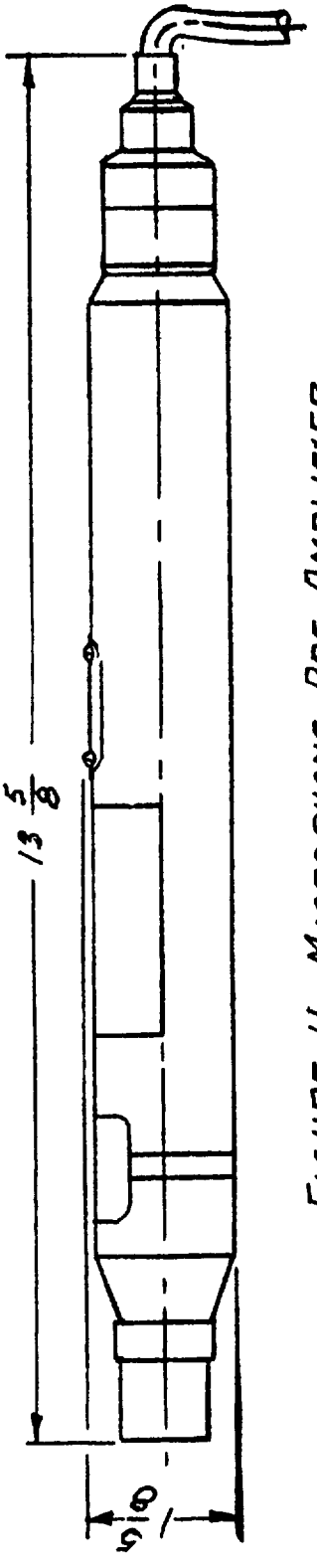


FIGURE 11. MICROPHONE PRE-AMPLIFIER

NOTE:
DIMENSIONS ARE IN INCHES.

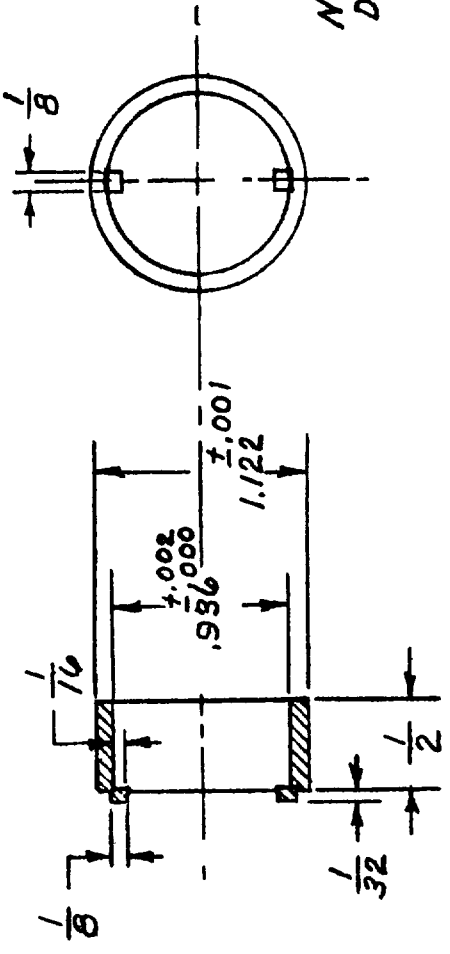


FIGURE 12. SIMULATED MICROPHONE CASE
(MATERIAL - BAKELITE)

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MILITARY SPECIFICATION

Microphone, Carbon, Oxygen Mask, ANB-M-C1

This Amendment forms a part of Military Specification MIL-M-22001(AER) dated 1 April 1959 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this Amendment basic MIL-M-22001(AER) is changed to MIL-M-22001(AS) and, throughout the specification, Bureau of Aeronautics shall be changed to Naval Air Systems Command.

Page 10, paragraph 6.2 Qualification, last sentence: Delete and substitute

> "Information pertaining to the qualification of products covered by this specification may be obtained from the Commanding Officer (Code QECA), U. S. Naval Ammunition Depot, Crane, Indiana 47522.

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